Cross-Domain Hyperspectral Target Detection Based on Physical Constraints and Spectral Generation

Zijin Li, Boya Zhao, *Member, IEEE*, Yuanfeng Wu, *Senior Member, IEEE*, and Bing Zhang, *Fellow, IEEE*

***Abstract*—To address the problems of significant spectral scale differences between multispectral images (MSIs) and hyperspectral images (HSIs), strong spectral variability of targets, and insufficient collaborative utilization of spatial–spectral features in cross-domain hyperspectral target detection (HTD), this article proposes a physically constrained spectral generation (SG) and dual-stream spatial–spectral fusion network for HTD (PSG-HTD). PSG-HTD comprises three parts. First, a cross-domain spectral self-attention (CSSA) mechanism is introduced to model long-range cross-band dependencies between MSIs and HSIs. Second, PSG-HTD incorporates a standard spectral library as a physical prior to improve the semantic rationality and physical credibility of generated spectra. Third, an adaptive dual-stream spatial–spectral extraction (ADSSE) module jointly exploits spectral discriminative and spatial structural features for cross-domain HTD. Experimental results demonstrate strong performance in both SG and cross-domain HTD tasks: 1) for SG, PSG-HTD reduces the root-mean-square error (RMSE) by up to 71.0% and improves the peak signal-to-noise ratio (PSNR) by about 37.4% over comparison methods, indicating that embedding physical constraints effectively alleviates the spectral distortion problem and ensures physical credibility and 2) for cross-domain HTD, it achieves the highest mAP@0.5 of 0.917 and 0.714 on RarePlanes and RSOD-VHR10, respectively, indicating that high-fidelity SG provides robust, domain-invariant features that significantly enhance downstream cross-domain target identification capability. This study provides a new idea for SG and cross-domain HTD, and has important application potential in the fields of fine recognition and accurate classification of remote sensing targets.**

***Index Terms*—Hyperspectral image (HSI), physical constraints, spatial–spectral feature extraction, spectral generation (SG).**

I. Introduction

THE acquisition methods of hyperspectral data have been diversified, covering various remote sensing platforms such as uncrewed aerial vehicles, aircraft, and satellites. The application of heterogeneous imaging sensors such as push-broom, whisk-broom, and staring types leads to significant scale differences in the representation of the same target in multisource image data. While enriching the observation dimensions, the diversity of data sources and the heterogeneity of imaging methods also bring challenges to the unified processing and information extraction of hyperspectral data.

Meanwhile, the practical application of hyperspectral technology still faces two major bottlenecks. First, acquiring and manually annotating hyperspectral images (HSIs) is costly, while sparse target distributions and category imbalance further aggravate the scarcity of high-quality annotated samples, limiting deep learning-based models. Second, the spatial and spectral resolutions of HSIs are constrained by the tradeoff imposed by optical diffraction and detector noise [1], [2]. Although HSIs can capture subtle spectral differences, insufficient spatial resolution may cause structural information loss for small targets, such as aircraft and vehicles, hindering fine recognition.

To alleviate the scarcity of annotated hyperspectral samples, label-rich multispectral remote sensing images provide a practical source for hyperspectral sample expansion. Spectral generation (SG) techniques, such as spectral super-resolution, HSI reconstruction, and HSI restoration, can recover HSI data from multispectral or RGB images [3]. This makes it possible to transfer abundant annotations from the multispectral domain to the hyperspectral domain, thereby providing data support for downstream cross-domain target detection. In this context, "target detection" in this article refers to object-level remote sensing target detection with bounding-box annotations, rather than traditional pixel-level spectral matching or anomaly detection. Correspondingly, "cross-domain" refers to the transfer from the label-rich RGB/multispectral image (MSI) source domain to the generated hyperspectral domain, where target detection is further performed based on the generated spatial–spectral representations.

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Zijin Li is with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: lizijin21@mails.ucas.ac.cn).

Boya Zhao and Yuanfeng Wu are with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: zhaoby@aircas.ac.cn; wuyf@aircas.ac.cn).

Bing Zhang is with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the College of Resources and Environment, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: zhangbing@aircas.ac.cn).

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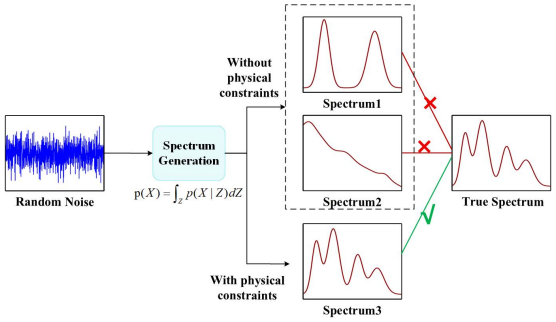


Fig. 1. Illustration of the physical inconsistency problem in existing data-driven SG methods.

HSI SG methods consist mainly of two categories: sparse dictionary learning-based and deep learning-based approaches. Early HSI SG methods mainly relied on sparse representation or dictionary learning to establish the mapping between low- and high-spectral-resolution data. Coupled dictionary learning [4] and joint sparse/low-rank learning [5] are representative examples of this category. These methods provide interpretable reconstruction frameworks, but their dependence on predefined sparse or low-rank assumptions limits their ability to model complex nonlinear spectral variations in remote sensing scenes.

Recent deep learning-based SG methods have improved reconstruction quality by introducing CNN and attention mechanisms. CNN-based methods, such as MIRNet [6], MPRNet [7] and HDNet [8], enhance image restoration or spectral reconstruction through multiscale residual learning, progressive restoration, or dual-domain modeling. However, these CNN-based methods mainly emphasize local spatial feature extraction or general image restoration, and their ability to explicitly model long-range cross-band dependencies remains limited. Transformer-based [9] methods further promote spectral reconstruction by introducing attention mechanisms. MST [10] and MST++ [11] model spectral dependencies through spectral-wise self-attention and multiresolution structures, while DSTrans [12] and VolFormer [13] enhance spatial–spectral interaction modeling through dual-stream spectral-spatial attention (SA) or volumetric attention. Although these attention-based methods effectively improve long-range dependency modeling, most of them are still driven mainly by pixel-level or band-level numerical reconstruction losses. They do not explicitly constrain the generated spectra according to standard spectral curves of real ground objects. Consequently, the generated results may appear visually reasonable but deviate from the physical reflectance characteristics of real spectra, such as abnormal spectral fluctuations, loss of absorption features, or distorted spectral morphology. This limitation weakens the physical credibility and interpretability of generated HSIs and may further reduce their reliability for downstream hyperspectral target detection (HTD) tasks.

As illustrated in Fig. 1, purely data-driven generation without physical constraints often yields physically distorted pseudo-spectra, either introducing abnormal fluctuations (Spectrum 1) or losing characteristic absorption features (Spectrum 2). Conversely, regularized by physical constraints, Spectrum 3 accurately preserves the global morphological trends and physical signatures of the true spectrum.

Apart from SG, cross-domain transfer learning has also been explored to alleviate the scarcity of labeled hyperspectral target samples. Existing HTD methods have attempted to construct transferable training samples or target–background representations from limited hyperspectral data. For example, CNNTD [14], CNND [15], and HTD-Net [16] improve HTD performance by generating target–background samples, pixel pairs, or augmented training data. However, these methods are mainly based on pixel-level sample construction within the hyperspectral domain, and their performance is often affected by additional manual labeling, reference scene completeness, background selection, or computational cost. More importantly, they do not fully exploit label-rich MSI remote sensing object detection datasets, nor do they explicitly model the cross-modal spectral transformation from MSI to HSI.

In summary, the practical problem considered in this work is the scarcity of large-scale HSI datasets with object-level annotations, despite the comparatively wide availability of label-rich MSI datasets. Addressing this problem requires overcoming three key challenges: the spectral-scale mismatch between MSI and HSI data, the insufficient physical credibility of generated spectra, and the insufficient joint exploitation of spatial and spectral information in cross-domain detection. To tackle the aforementioned issues, this article proposes a physically constrained SG and dual-stream spatial–spectral fusion for the HTD network (PSG-HTD), with the main contributions as follows.

1) A cross-domain spectral self-attention (CSSA) mechanism is proposed to alleviate the spectral-scale mismatch between MSI and HSI representations. By introducing learnable spectral reweighting into self-attention, CSSA adaptively models long-range cross-band dependencies across different spectral densities and improves cross-domain spectral mapping.
2) A physically constrained SG framework is developed to improve the reliability of generated HSI. Standard ground-object spectral libraries are introduced as physical priors, together with a spectral angle mapper (SAM) constraint, to encourage the generated spectra to preserve realistic spectral characteristics and improve their physical credibility.
3) An adaptive dual-stream spatial–spectral extraction (ADSSE) module is introduced to effectively exploit the complementary spectral and spatial information of generated HSI for cross-domain target detection. ADSSE separately extracts principal component analysis (PCA)-compressed spectral signatures and optimal neighborhood reconstruction (ONR)-selected spatial structures, and performs bidirectional interaction through cross-attention to learn joint representations with enhanced discriminability and domain invariance.

The remainder of this article is organized as follows. Section II reviews existing work on HSI SG and target detection. Section III details our proposed cross-domain HTD algorithm. Section IV introduces and analyzes the

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experimental results, including comparisons with existing state-of-the-art methods. Section V concludes this article.

II. Related Work

A. SG for HSIs

Existing SG methods have evolved from traditional interpolation and sparse recovery techniques to deep learning approaches, and can generally be categorized into prior knowledge-based and deep learning-based methods.

Prior knowledge-based methods exploit image characteristics such as sparsity [17], spatial structures [18], and spectral correlations [19] for reconstruction. However, such priors mainly improve numerical reconstruction or statistical consistency, and do not explicitly constrain generated spectra according to standard spectral curves of real ground objects. Therefore, physically implausible spectra may still be generated when reconstructed curves satisfy numerical fitting criteria but deviate from real spectral morphology.

Deep learning has significantly advanced hyperspectral SG through end-to-end learning without manual feature engineering. Existing methods mainly include Unet structures, residual learning, GANs, attention mechanisms, and Transformer-based architectures [3]. Unet-based methods [20], [21] effectively capture multiscale features, while residual learning [22], [23], [24] improves reconstruction stability and high-frequency detail recovery. GAN-based methods [25] further enhance spectral realism, and attention mechanisms [26], [27] strengthen spatial–spectral feature representation. More recently, Transformer-based methods [10], [11], [12], [13], [28] have demonstrated superior capability in modeling long-range spectral-spatial dependencies. Nevertheless, most of these methods still optimize pixel-wise or band-wise reconstruction errors and lack explicit material-level physical constraints. As a result, generated HSIs may achieve high numerical reconstruction accuracy but still exhibit abnormal reflectance ranges, distorted spectral morphology, or missing characteristic peaks and valleys. This limitation reduces their physical credibility and weakens their reliability for downstream HTD.

B. HTD Based on Cross-Domain Transfer Learning

In HTD, target samples are usually sparse and manually annotated HSI datasets are difficult to obtain. To alleviate this problem, domain adaptation methods [29] have been introduced to transfer knowledge from label-rich source domains to label-scarce target domains. Domain-adaptive target detection techniques are typically divided based on the level of adaptation into instance-level, feature-level, and model parameter-level categories [30], [31].

Instance-level methods reduce the domain gap by modifying input appearance, such as image-to-image translation [32], domain randomization [33], [34], and style transfer [35], [36]. Although these methods can reduce low-level appearance differences, model performance shows strong dependence on the quality of generated instances. Feature-level methods align feature distributions between source and target domains through adversarial learning, teacher–student frameworks, metric constraints, or semantic alignment [37], [38], [39], [40], [41], [42]. Model parameter-level methods further improve adaptation by pseudo-label refinement, meta-learning, or task-specific knowledge transfer [43], [44], [45], [46], [47]. Although these approaches have achieved progress in RGB or multispectral object detection, they mainly address appearance-level or feature-distribution shifts. They do not explicitly model the spectral-domain discrepancy between MSI/RGB and HSI data, which is essential for HTD.

For HTD and analysis, transfer learning has also been investigated to alleviate the scarcity of labeled samples. Existing studies have explored knowledge transfer for HSI analysis [48], unsupervised transfer learning for HTD [49], feature adaptation and augmentation for cross-scene HSI classification [50], transfer learning-based HTD [51], semi-supervised domain-adaptive few-shot learning [52], RGB-to-HSI spatial feature transfer for large-scale target detection [53], spectral diffusion for hyperspectral anomaly detection [54], and Transformer-based cross-domain few-shot learning [55]. These studies demonstrate the potential of transfer learning for HTD. However, most of them focus on same-modality transfer among different HSI scenes or sensors, or on classification/anomaly detection settings. The problem of generating physically credible HSI data from label-rich RGB/MSI detection datasets and further performing object-level HTD remains insufficiently explored.

Compared with existing cross-domain detection and HTD methods, PSG-HTD focuses on a spectral-generation-assisted transfer setting. It first uses label-rich RGB/MSI remote sensing data as the source domain and generates physically credible hyperspectral data through spectral library constraints. Then, ADSSE extracts and fuses spectral and spatial features from the generated HSI to enhance object-level target detection. Therefore, the proposed framework simultaneously addresses the shortage of annotated HSI detection samples, the physical credibility of generated spectra, and the insufficient spatial–spectral fusion in cross-domain HTD.

III. Method

Fig. 2 presents the proposed PSG-HTD framework. As depicted, PSG-HTD adopts a cascaded end-to-end architecture of SG-based HTD, embedding the SG module based on spectral library physical constraints as the core carrier of sample expansion and cross-domain knowledge transfer into the adaptive dual-stream spatial–spectral feature target detection network, realizing cross-domain feature transfer and high-precision target detection from the multispectral source domain to the hyperspectral target domain.

The PSG-HTD comprises two stages. The first stage is high-credibility SG based on spectral library physical constraints. Given label-rich MSIs as input, it employs CSSA with learnable spectral reweighting coefficients to adaptively model cross-band dependencies and alleviate spectral scale mismatch during cross-domain spectral mapping. Meanwhile, a standard ground-object spectral library is introduced as a physical prior, and a SAM-based consistency loss is constructed to constrain the generated spectra. In this way, physically reliable HSIs are

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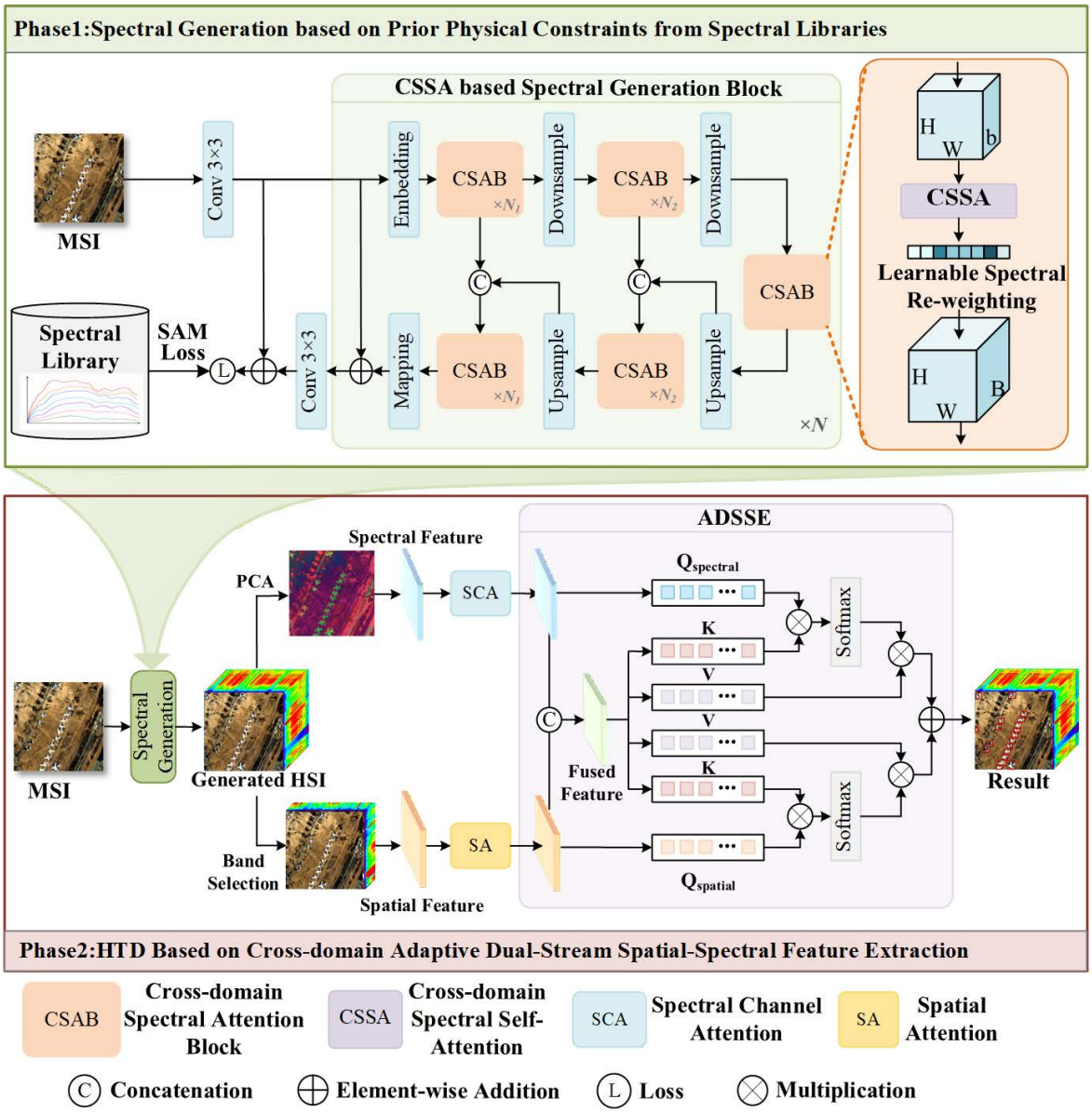


Fig. 2. Flowchart of the proposed PSG-HTD.

generated, enabling the expansion of annotated hyperspectral samples and preliminary cross-domain spectral alignment. The second stage is cross-domain HTD based on adaptive dual-stream spatial–spectral feature extraction. The generated HSIs are decomposed into spectral and spatial representations through PCA and band selection, respectively. ADSSE is then employed to achieve bidirectional spatial–spectral guidance and domain-invariant joint representation learning. Finally, the fused representation is fed into the detection network to achieve high-precision object-level HTD under cross-domain scenarios.

A. Cross-Domain Spectral Self-Attention

The spectral-scale mismatch between sparse multispectral bands and dense hyperspectral bands poses a challenge to cross-domain spectral mapping. To alleviate this mismatch, CSSA models long-range cross-band dependencies through learnable spectral reweighting, enabling adaptive spectral feature alignment for hyperspectral generation.

Given the mathematical representation of the source-domain MSI as $\mathbf{X}_{MSI} \in \mathbb{R}^{H\times W\times b}$, where H and W denote the image height and width, and b denotes the band number of the MSI. The expected output of the target domain HSI is $\mathbf{X}_{HSI} \in \mathbb{R}^{H\times W\times B}$, where B denotes the number of bands in the HSI. The standard spectral library is $\mathbf{S} = \{\mathbf{s}_k\}_{k=1}^{K}$, where $\mathbf{s}_k \in \mathbb{R}^B$ is the standard spectral vector for the kth class of typical targets, and K is the total number of target categories in the spectral library. SG seeks to learn a nonlinear mapping under spectral library physical constraints, such that the generated HSI $\hat{\mathbf{X}}_{HSI}$ satisfies both consistency with the input MSI and physical consistency with the standard spectral library.

To achieve this goal, a U-shaped SG Transformer block (SGTB) is designed as the SG unit. The SG network is designed end-to-end by sequentially connecting N SGTBs, allowing progressive refinement of spectral features from

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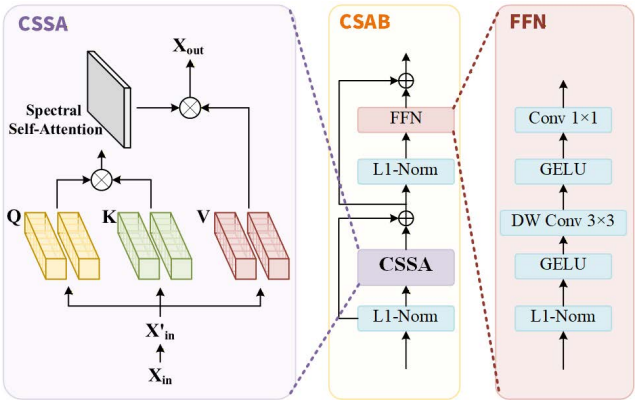


Fig. 3. Illustration of CSAB module.

coarse to fine. The upper portion of Fig. 2 presents the proposed SG framework in detail. First, a 3 × 3 convolutional layer is employed to derive shallow feature maps $\mathbf{F}_{MSI} \in \mathbb{R}^{H\times W\times l}$ from the multispectral input, where l is the channel number for shallow features. The shallow representation is subsequently fed into SGTB blocks for deep feature extraction. The SGTB adopts an encoder-bottleneck-decoder U-shaped structure. Its core design idea is to compress spatial dimensions through downsampling to strengthen spectral feature expression, then restore spatial resolution through upsampling, while preserving shallow detail information via skip connections. The encoder consists of two downsampling operations and $\mathbf{N}_1 + \mathbf{N}_2$ cross-domain spectral attention blocks (CSABs). The bottleneck layer is composed of $\mathbf{N}_3$ CSABs connected in series. The decoder adopts a structure symmetric to the encoder, using transposed convolution for upsampling. A transposed convolution with a 3 × 3 kernel, stride 2, and padding 1 is adopted to ensure that the upsampled feature map has the same spatial size as the corresponding encoder stage. To compensate for detail loss during downsampling, skip connections are introduced between the matching decoder and encoder layers. Feature dimensions are adjusted via 1 × 1 convolution before concatenation. The embedding and mapping layers both use 3 × 3 convolutional layers. The CSAB is the fundamental unit for spectral feature extraction in the network. Its architecture, depicted in Fig. 3, comprises CSSA, a feed-forward network (FFN), and two LayerNorm layers. It employs prenormalization design (i.e., normalization before attention calculation and feed-forward operation), effectively alleviating gradient vanishing issues during training.

Fig. 3 depicts the detailed design of CSSA. The input feature $\mathbf{X}_{in} \in \mathbb{R}^{H\times W\times C}$ is flattened along the spatial dimension, transforming its shape into 2-D token $\mathbf{X}'_{in} \in \mathbb{R}^{(H\times W)\times C}$. Subsequently, a standard linear projection is applied to $\mathbf{X}'_{in}$ to obtain the Query $\mathbf{Q}$, Key $\mathbf{K}$, and Value $\mathbf{V}$. To enhance spectral dependency modeling, the obtained $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ features are further divided into N parallel spectral branches: $\mathbf{Q} = \{\mathbf{Q}_n\}_{n=1}^N$, $\mathbf{K} = \{\mathbf{K}_n\}_{n=1}^N$ and $\mathbf{V} = \{\mathbf{V}_n\}_{n=1}^N$, where $\mathbf{Q}_n, \mathbf{K}_n, \mathbf{V}_n \in \mathbb{R}^{(H\times W)\times d}$ and dimension $d = C/N$. In CSSA, each spectral representation is considered as a token, and self-attention is calculated for $\mathbf{head}_n$

$$\mathbf{A}_n = \text{Softmax}\left(\boldsymbol{\varepsilon}_n \mathbf{Q}_n^{\mathrm{T}} \mathbf{K}_n\right), \mathbf{head}_n = \mathbf{V}_n \mathbf{A}_n \tag{1}$$

where $\boldsymbol{\varepsilon}_n$ is a learnable parameter for adjusting weights of self-attention $\mathbf{A}_n$. Unlike conventional self-attention that treats all token interactions through a fixed attention formulation, this parameter dynamically adjusts the attention contribution of different spectral bands by reweighting the matrix multiplication $\mathbf{Q}_n^{\mathrm{T}}\mathbf{K}_n$ within $\mathbf{head}_n$, addressing the issue of varying spectral density with wavelength, thereby better modeling cross-domain band mapping relationships. The outputs from all attention heads are concatenated, and the resulting feature is projected linearly to obtain the final attention representation

$$\mathbf{CSSA}(\mathbf{X}) = \left(\underset{n=1}{\overset{N}{\text{Concat}}}(\mathbf{head}_n)\right)\mathbf{W}_0 + \mathbf{b}_0 \tag{2}$$

where $\mathbf{W}_0 \in \mathbb{R}^{C\times C}$ denotes the output projection matrix and $\mathbf{b}_0$ denotes the bias term. Finally, the result from the above operations is reshaped to obtain the output feature $\mathbf{X}_{out} \in \mathbb{R}^{H\times W\times C}$.

Thus, CSSA provides cross-band dependency modeling and adaptive spectral feature alignment for physically credible generation and downstream cross-domain HTD.

B. SG Based on Prior Physical Constraints From Spectral Libraries

Data-driven SG alone may produce spectra inconsistent with the reflectance characteristics of real ground objects. Building on CSSA in Section III-A, this module incorporates physical priors from a standard spectral library through a spectral consistency loss to improve the physical credibility of generated spectra.

As the core carrier of sample expansion and cross-domain knowledge transfer in PSG-HTD, the input of this module is MSIs with rich labels, and the output is HSIs conforming to the physical laws of real ground objects. The SG under the constraint of spectral library can be regarded as an optimization problem to solve

$$H = \underset{H}{\text{argmin}} \|\mathbf{M} - \boldsymbol{\Phi}(\mathbf{H})\|^2 + \lambda * \boldsymbol{\psi}(\mathbf{H}, \mathbf{S}) \tag{3}$$

where $\mathbf{H}$ denotes the HSI to be generated, $\mathbf{M}$ is the input MSI, $\boldsymbol{\Phi}$ represents the spectral dimensionality reduction from HSI to MSI, and $\boldsymbol{\psi}(\mathbf{H}, \mathbf{S})$ serves as a regularization term that enforces the generated hyperspectral spectra to align with the standard spectral curves in the library $\mathbf{S}$. The optimization objective is to learn a nonlinear mapping that preserves consistency with the input MSI while ensuring the physical reliability of the generated hyperspectral spectra.

Standard spectral libraries typically contain spectral data for hundreds of target categories. However, specific remote sensing scenes (e.g., airports, farmland, and ports) often focus on only a few typical target categories (e.g., plane, vegetation, water bodies, and buildings). Directly using the global spectral library for constraints would introduce irrelevant information interference, reducing model training efficiency and constraint accuracy. Therefore, this article takes the airport scene, which is a key focus in remote sensing, as an example. First, typical spectra of interest are selected from existing spectral libraries, covering main target categories (e.g., plane) and background

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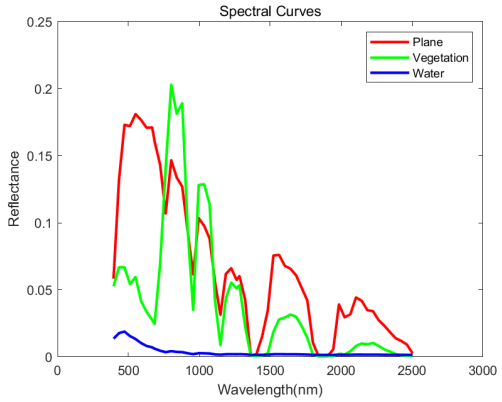


Fig. 4. Spectral curves of typical objects in AVIRIS remote sensing images.

categories (e.g., vegetation and water bodies). Fig. 4 depicts the selected typical spectral curves.

Denote the set of endmember ground-truth spectra from the selected spectral library as $\mathbf{S}_0 = \{\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3\}$, where each endmember spectrum $\mathbf{s}_i \in \mathbb{R}^B$ is a 1-D vector, and B denotes the spectral band number.

To comprehensively constrain the physical plausibility of generated spectra, this article constructs a multidimensional loss function, including SAM loss and image reconstruction loss, co-optimizing the model. Different from pixel-wise numerical reconstruction metrics such as root-mean-square error (RMSE), peak signal-to-noise ratio (PSNR), and RAE, which mainly measure the absolute reconstruction error between generated and reference images, SAM measures the angular similarity between two spectral vectors. Therefore, SAM can more directly reflect whether the generated spectrum preserves the intrinsic spectral morphology of real ground objects. A smaller SAM value indicates that the generated spectrum is more consistent with the reference spectrum in terms of spectral shape and reflectance trend. During SG, it is necessary to constrain that the spectral curves of pixels belonging to various object categories in the generated HSI should be as consistent as possible with the corresponding ground-truth spectra according to their semantic categories. Therefore, a SAM loss term $\mathcal{L}_{\text{SAM}}$ is constructed to reduce the discrepancy between generated and ground-truth spectra

$$\mathcal{L}_{\text{SAM}} = \frac{1}{HW} \times \sum_{i=1}^{H}\sum_{j=1}^{W} \min_{\mathbf{s_k}} \left(\arccos \left(\frac{\hat{\mathbf{X}}_{HSI}(i,j,:) \cdot \mathbf{s}_k^T}{\left\|\hat{\mathbf{X}}_{HSI}(i,j,:)\right\|_2 \cdot \|\mathbf{s}_k\|_2} \right) \right) \tag{4}$$

where $\mathbf{s_k} \in \mathbf{S_0}$ is the ground-truth spectra from the spectral library, and $\hat{\mathbf{X}}_{HSI}(i,j,:)$ is the generated spectra. For each spatial location (i, j) of each object category in the generated HSI $\hat{\mathbf{X}}_{HSI}$, the SAM between its generated spectral vector $\hat{\mathbf{X}}_{HSI}(i,j,:)$ and all endmember spectra in the spectral library $\mathbf{S}_0$ is calculated, and the minimum value is taken as the SAM loss for that pixel. This loss guides the generated spectra to align with the ground-truth spectral curves, thereby avoiding spectral distortions.

Additionally, to ensure consistency between the generated HSIs and the real HSIs, the RMSE loss is applied to evaluate the difference between $\hat{\mathbf{X}}_{HSI}$ and $\mathbf{X}_{HSI}$ (5), as shown at the bottom of the page.

This loss term $\mathcal{L}_{\text{RMSE}}$ and the spectral loss $\mathcal{L}_{\text{SAM}}$ together constitute the total loss function

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{RMSE}} + \lambda \mathcal{L}_{\text{SAM}} \tag{6}$$

where λ is a weighting hyperparameter. By explicitly introducing a physical-consistency loss, this module goes beyond purely data-driven SG and integrates physical priors from a standard spectral library into the generation process. The generated HSIs therefore exhibit both high visual quality and physical credibility, providing reliable training samples for the subsequent adaptive dual-stream spatial–spectral detection network while facilitating preliminary cross-domain knowledge transfer from multispectral to hyperspectral data.

C. HTD Based on Cross-Domain Adaptive Dual-Stream Spatial–Spectral Feature Extraction

Following SG, effective cross-domain HTD requires joint exploitation of spatial–spectral information while mitigating domain shift. Accordingly, a cross-domain ADSSE module is developed, with parallel spectral and spatial streams and bidirectional feature fusion to learn domain-invariant joint representations for target detection.

1) Spectral Stream Feature Extraction: The goal of the spectral extraction branch is to capture discriminative spectral features of targets in the HSI. The specific process is as follows.

1) *Dimensionality Reduction by PCA:* Perform pixel-level PCA dimensionality reduction on the generated HSI $\hat{\mathbf{X}}_{HSI}$, retaining the first K principal components to generate the spectral feature matrix $\mathbf{F}_{\text{spec}} \in \mathbb{R}^{H\times W\times K}$. PCA minimizes redundancy along the spectral dimension, while preserving the most discriminative spectral features and lowering computational complexity for subsequent steps.

2) *Spectral Channel Attention (SCA) Enhancement:* As illustrated in Fig. 5, to enhance the feature contribution of key spectral bands, an SCA module is introduced. This module models the importance of different spectral channels, assigning adaptive weights to each principal component. First, global average pooling is applied to $\mathbf{F}_{\text{spec}}$ to produce the channel feature vector $\mathbf{F}_{\text{spec}}^{\text{avg}} \in \mathbb{R}^K$. Then, the channel attention weights $\mathbf{W}_{\text{spec}}$ are derived using a two-layer fully connected network with a Sigmoid activation function. The attention weights are subsequently applied to the original spectral features to yield the enhanced spectral features $\mathbf{F}'_{\text{spec}} = \mathbf{F}_{\text{spec}} \cdot \mathbf{W}_{\text{spec}}$.

$$\mathcal{L}_{\text{RMSE}} = \sqrt{\frac{1}{HWB}\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{k=1}^{B}\left[\hat{\mathbf{X}}_{HSI}(i,j,k) - \mathbf{X}_{HSI}(i,j,k)\right]^2}. \tag{5}$$

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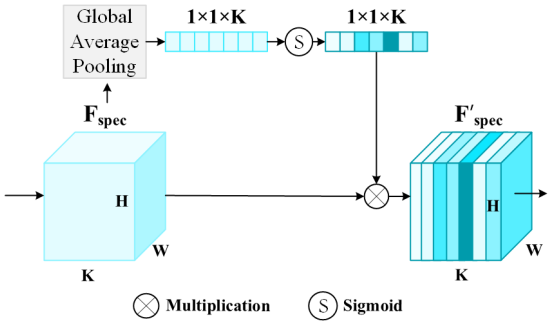


Fig. 5. Architecture of SCA.

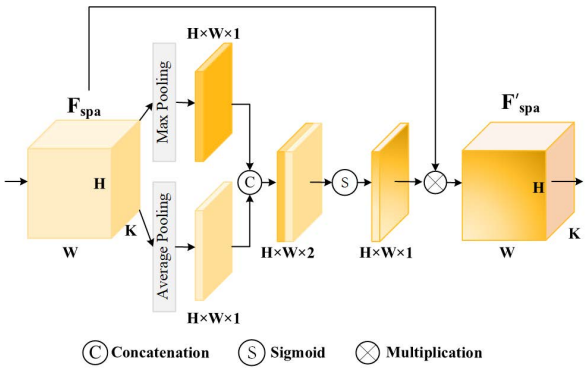


Fig. 6. Architecture of SA.

2) Spatial Stream Feature Extraction: The spatial extraction branch aims to capture spatial structural features (e.g., shape, edge, and texture) of targets from the HSI. The specific process is as follows.

1) *Band Selection Based on ONR Algorithm [56]:* Adjacent spectral bands in HSIs are highly correlated, and some bands may have more noise, causing redundancy and interference in spatial data. The ONR algorithm can select bands that are more representative of spatial information and less affected by noise. For each band $\hat{\mathbf{X}}_{HSI}(:,:,k)$ of $\hat{\mathbf{X}}_{HSI}$, the finally selected bands should be those with minimal reconstruction error. The top K bands minimizing reconstruction error are selected to form the spatial feature input $\mathbf{F}_{\text{spa}} \in \mathbb{R}^{H\times W\times K}$.

2) *SA Enhancement:* As depicted in Fig. 6, to enhance spatial feature representation in target regions, the SA module is incorporated. First, average pooling and max pooling along the channel dimension are performed on $\mathbf{F}_{\text{spa}}$ to obtain corresponding spatial feature maps $\mathbf{F}_{\text{spa}}^{\text{avg}} \in \mathbb{R}^{H\times W}$ and $\mathbf{F}_{\text{spa}}^{\max} \in \mathbb{R}^{H\times W}$, which are concatenated to get $\mathbf{F}_{\text{spa}}^{\text{cat}} \in \mathbb{R}^{H\times W\times 2}$. Then, the SA weights $\mathbf{w}_{\text{spa}}$ are computed via a Sigmoid activation. Finally, the SA weights are multiplied with the original spatial features to yield the enhanced spatial features $\mathbf{F}'_{\text{spa}} = \mathbf{F}_{\text{spa}} \cdot \mathbf{W}_{\text{spa}}$.

3) Cross-Domain Adaptive Dual-Stream Fusion Module: The ADSSE module is designed to facilitate spatial–spectral feature fusion and cross-domain distribution alignment, addressing the domain shift problem. To achieve bidirectional guided fusion of spectral and spatial stream features, this module employs a cross-attention mechanism. First, the previously obtained spectral features and spatial features are encoded to get $\mathbf{F}''_{\text{spec}} \in \mathbb{R}^{HW\times d_k}$ and $\mathbf{F}''_{\text{spa}} \in \mathbb{R}^{HW\times d_k}$. To aggregate contextual information from image spatial–spectral features, the two image features are concatenated and further fused as the final spatial–spectral representation $\mathbf{F}_{\text{fused}}$. The spectral fusion cross-attention is then calculated by using the spectral features as query (Q) and the fused features as key (K) and value (V)

$$\mathbf{Attn_1} = \text{Softmax}\left(\frac{\mathbf{F}''_{\text{spec}}\mathbf{F}_{\text{fused}}^{T}}{\sqrt{d_k}}\right)\mathbf{F}_{\text{fused}}. \tag{7}$$

Simultaneously, using the spatial features as query (Q) and the fused features as key (K) and value (V), the spatial fusion cross-attention is calculated

$$\mathbf{Attn_2} = \text{Softmax}\left(\frac{\mathbf{F}''_{\text{spa}}\mathbf{F}_{\text{fused}}^{T}}{\sqrt{d_k}}\right)\mathbf{F}_{\text{fused}}. \tag{8}$$

Finally, a 1×1 convolution is applied to the summed cross-attention features to adjust the channel dimension, producing the fused feature $\mathbf{F}'_{\text{fused}}$. By enabling mutual guidance between spatial and spectral features, the ADSSE enhances their joint representation. The resulting dual-stream framework jointly exploits spatial–spectral information and aligns source-target feature distributions, thereby improving cross-domain HTD accuracy and generalization.

IV. Experiments

A. Datasets

The datasets consist of two parts: datasets for SG and datasets for HTD. Considering the inherent tradeoff between spatial and spectral resolutions in hyperspectral imaging, the limited spectral resolution of RGB/MSI data, and the lack of large-scale annotated HTD datasets, we construct the HTD evaluation data in a controlled cross-domain manner. Specifically, label-rich high-spatial-resolution RGB/MSI remote sensing object detection datasets are first used as the source-domain data. These images are spatially aligned with HSIs in terms of spatial resolution through downsampling. Then, the physically constrained SG module is applied to enhance their spectral resolution and generate label-aligned hyperspectral data under spectral library priors. The generated hyperspectral data inherit the object-level annotations from the source datasets and are further used for downstream target detection. This protocol enables the evaluation of spectral-generation-assisted cross-domain HTD under the current condition where large-scale real annotated hyperspectral detection datasets are difficult to obtain.

1) Datasets for SG: For this study, a paired Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) multispectral–hyperspectral dataset AMH was collected and annotated. The hyperspectral data comes from several full-scene HSIs captured by the AVIRIS sensor, with an original band number of 224 and spatial resolution ranging between 0.9 and 7.3 m. After manually removing some cloudy or overexposed data, only bands 1:4:224 from the original 224 bands were retained, compressing it to 56 bands. For each HSI, its data file simultaneously contains the hyperspectral 3-D cube data, the corresponding multispectral data, and the corresponding object ground truth (containing ground truth for three types of objects: aircraft, vegetation, water bodies).

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Fig. 7. f170127t01p00r08rdn_e data. (a) False color image of MSI, (b) ground truth of plane, (c) ground truth of vegetation, and (d) ground truth of water.

The ground truth comes from manual annotation and is used to calculate the separate SAM loss for each object type. Taking one precropped full-scene AVIRIS hyperspectral data, f170127t01p00r08rdn_e, as an example, the corresponding MSI (false color) and ground truth are shown in Fig. 7.

To facilitate network training, images were uniformly cropped to a size of 400 × 400 × 56 hyperspectral data. All HSIs have paired MSIs, and all underwent image normalization processing. Finally, the dataset comprises 387 multispectral–hyperspectral pairs, divided into 344 training pairs and 43 validation pairs.

2) Datasets for HTD:

a) RarePlanes dataset: The RarePlanes dataset is a remote sensing RGB image dataset with high spatial resolution, including multiple satellite images from Maxar and AI.Reverie, respectively. Among them, there are 253 Maxar WorldView-3 satellite images from 112 locations, with about 14 700 manually annotated aircraft. The remaining data comes from the AI.Reverie platform, including 50 000 satellite images and more than 600 000 aircraft labels. The dataset comprises a total of 8525 images of size 512 × 512, split into 6820 training images and 1705 test images following an 8:2 ratio. To better simulate the morphological characteristics of targets in HSIs with limited spatial resolution, the dataset is uniformly downsampled by a factor of 4. Based on these downsampled RGB images, physically constrained SG is further performed to obtain generated hyperspectral data for downstream target detection.

b) RSOD-VHR10 dataset: The RSOD-VHR10 dataset is formed from two public remote sensing target detection datasets, RSOD [57] and NWPU VHR-10 [58], [59]. It includes images covering the plane category from RSOD and all images from NWPU VHR-10, retaining only the plane category labels, totaling 1246 images. The RSOD dataset mainly comes from Google Earth, Google Maps, and other public datasets, with image resolutions ranging between 0.5 and 2 m. The NWPU VHR-10 dataset mainly comes from Google Earth with resolutions from 0.5 to 2 m, and image sizes ranging from 500 × 500 to 1100 × 1100 pixels. After dataset division, the training set has 996 images, and the test set has 250 images. Following the same protocol as RarePlanes, the RGB images in RSOD-VHR10 are uniformly downsampled by a factor of 4 to simulate the lower spatial resolution commonly observed in hyperspectral imaging.

B. Experimental Setting

We selected the Yolov5 network as the backbone for the cross-domain dual-stream spatial–spectral feature extraction network. Experiments were performed on the RarePlanes and RSOD-VHR10 datasets to evaluate the effectiveness of the proposed cross-domain HTD method. All compared detectors are evaluated under the same dataset split and evaluation metrics. This setting is designed to verify whether SG can compensate for the limited spectral resolution of RGB data and improve target detection under a low-spatial-resolution hyperspectral scenario. The network was implemented based on the PyTorch framework, trained on four NVIDIA A6000 GPUs (48 GB). The initial learning rate was set to 0.01, with a batch_size of 16, and all experiments were trained for 100 epochs. For HTD algorithm comparison and evaluation, precision, recall, mAP@0.5, and mAP@0.5:0.95 were chosen as evaluation metrics. For the physically constrained SG network, we chose the Transformer framework and trained and validated it on the AMH dataset.

C. Results and Analysis

We separately evaluate SG quality and cross-domain target detection performance to verify the physical credibility of generated spectra and the effectiveness of PSG-HTD.

1) Comparative Experiments of SG Methods: For SG, plane, vegetation, and water spectra from the library are used as physical constraints. We compare the proposed SG method with MST [10], HINet [60], MIRNet [6], HDNet [8], Restormer [61], and MPRNet [7] on AMH using RMSE, RAE, PSNR, global SAM, and category-specific SAM. Results are reported in Table I.

In Table I, lower RMSE, RAE, and SAM and higher PSNR indicate better performance. For better comparison, all metrics were calculated after image normalization. The proposed method achieves the best RMSE (0.0126), RAE (0.1625), and PSNR (38.7930), demonstrating strong numerical reconstruction. Its global SAM (0.1738) is slightly higher than MST and Restormer, but it achieves the best vegetation SAM (0.0443), while the plane and water SAM values remain low. These results indicate that the joint reconstruction and physical-prior constraints provide a balanced reconstruction across numerical fidelity and material-specific spectral consistency rather than optimizing a single metric.

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TABLE I
COMPARISON OF GENERATION RESULTS FROM DIFFERENT SG METHODS

Method	RMSE	RAE	PSNR	SAM	Plane_SAM	Veg_SAM	Water_SAM
MST	0.0209	0.3538	34.2160	0.1650	0.0081	0.0540	0.0759
HINet	0.0419	0.8156	28.2735	0.3621	0.0099	0.1751	0.2201
MIRNet	0.0434	0.7466	28.2273	0.4062	0.0100	0.1977	0.1864
HDNet	0.0167	0.2814	36.5901	0.1855	**0.0039**	0.0689	0.0876
Restormer	0.0164	0.3411	36.8060	**0.1453**	0.0048	0.0564	0.1694
MPRNet	0.0221	0.6707	34.9429	0.2048	0.0091	0.0482	**0.0593**
Proposed	**0.0126**	**0.1625**	**38.7930**	0.1738	0.0096	**0.0443**	0.0845

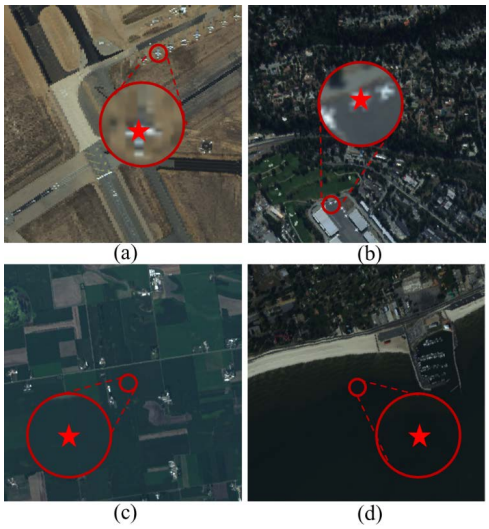


Fig. 8. Sample test images for SG. (a) and (b) Plane samples. (c) Vegetation sample. (d) Water sample.

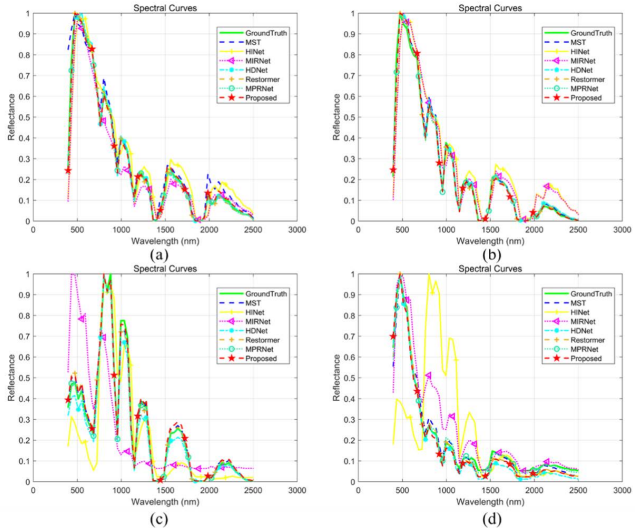


Fig. 9. Comparison of generated spectral curves of different SG methods on three types of ground objects. (a) and (b) Plane spectra (c) Vegetation spectra (d) Water spectra. (Note: all spectral curves are normalized.)

To further assess physical credibility, Fig. 8 shows representative samples used for spectral-curve comparison, where (a) and (b) represent planes, (c) represents vegetation, and (d) represents water bodies.

Fig. 9 compares generated and ground-truth spectra at the marked pixels. All spectral curves are normalized for easy comparison. The proposed method closely follows the overall spectral shapes of planes, vegetation, and water. HINet and MIRNet show larger deviations; MST deviates near some plane peaks and water bands, while HDNet shows mismatches for vegetation and water at specific wavelengths. Restormer and MPRNet are closer in curve shape but have inferior RMSE, RAE, and PSNR. Together with Table I, the visual comparison confirms that the proposed method improves spectral credibility while maintaining strong reconstruction accuracy, providing reliable data support for the subsequent downstream HTD task.

2) Comparative Experiments of Cross-Domain HTD Algorithms: We compared the PSG-HTD with several state-of-the-art target detection algorithms, including Faster RCNN [62], RetinaNet [63], Cascade RCNN [64], FCOS [65], DETR [66], Sparse RCNN [67], DINO [68], Yolov8, and Yolov11, as well as three representative deep learning-based hyperspectral object detectors, DoubleFPN [69], SpecDETR [70], and DAHNet [71]. To better simulate the spatial-resolution characteristics of hyperspectral imagery, all images were uniformly downsampled by a factor of 4. For a fair comparison, the conventional RGB detectors directly use the corresponding downsampled RGB images as input, whereas DoubleFPN, SpecDETR, DAHNet, and PSG-HTD are evaluated using hyperspectral data generated from the corresponding RGB images using the proposed physically constrained SG method under the same spatial-resolution setting. For both the spectral and spatial streams, $K = 3$ was adopted, preserving over 99.4% of the cumulative variance while keeping the ONR reconstruction error below 0.05 on both datasets, providing a good tradeoff between information preservation and computational efficiency.

1) *Validation Results on the RarePlanes Dataset:*

Table II shows the accuracy comparison of different target detection algorithms on the RarePlanes dataset. The table indicates that the PSG-HTD target detection algorithm achieved the highest precision, recall, and mAP@0.5 on the RarePlanes dataset, with values of 0.923, 0.861, and 0.917, respectively, demonstrating its stronger ability to reliably detect more target instances while maintaining high detection precision. Although SpecDETR achieves a slightly higher mAP@0.5:0.95, the superior results of PSG-HTD on the other three metrics indicate better overall detection performance.

Figs. 10 and 11 compare detection results on two representative RarePlanes images, where yellow boxes indicate ground-truth labels and the confidence threshold is set to conf_thre = 0.5. The images show aircraft on an apron

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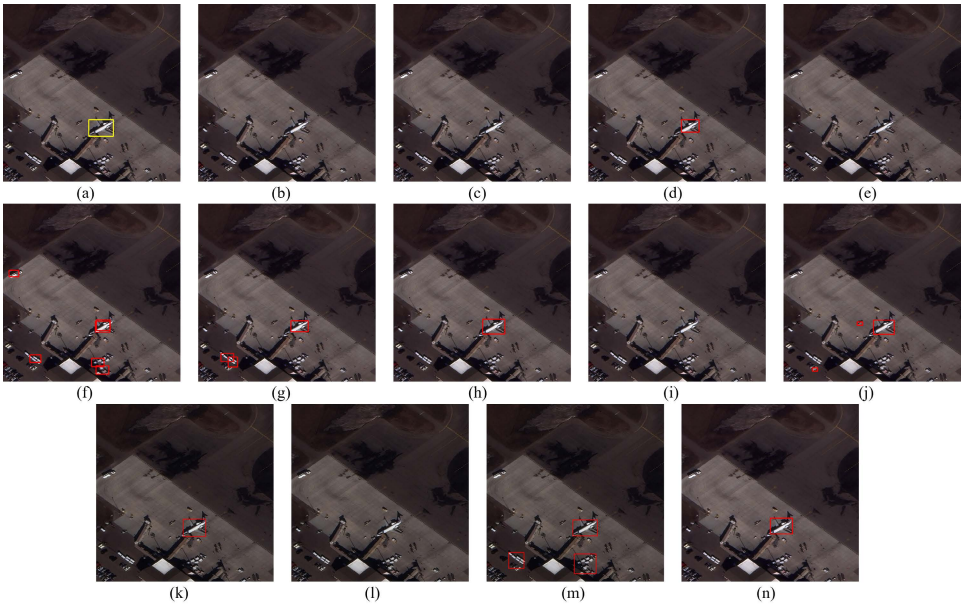


Fig. 10. Detection results of different object detection algorithms on example image 1 from the RarePlanes dataset (conf_thre = 0.5). (a) GT. (b) Faster RCNN. (c) RetinaNet. (d) Cascade RCNN. (e) FCOS. (f) DETR. (g) Sparse RCNN. (h) DINO. (i) Yolov8. (j) Yolov11. (k) DoubleFPN. (l) SpecDETR. (m) DAHNet. (n) PSG-HTD.

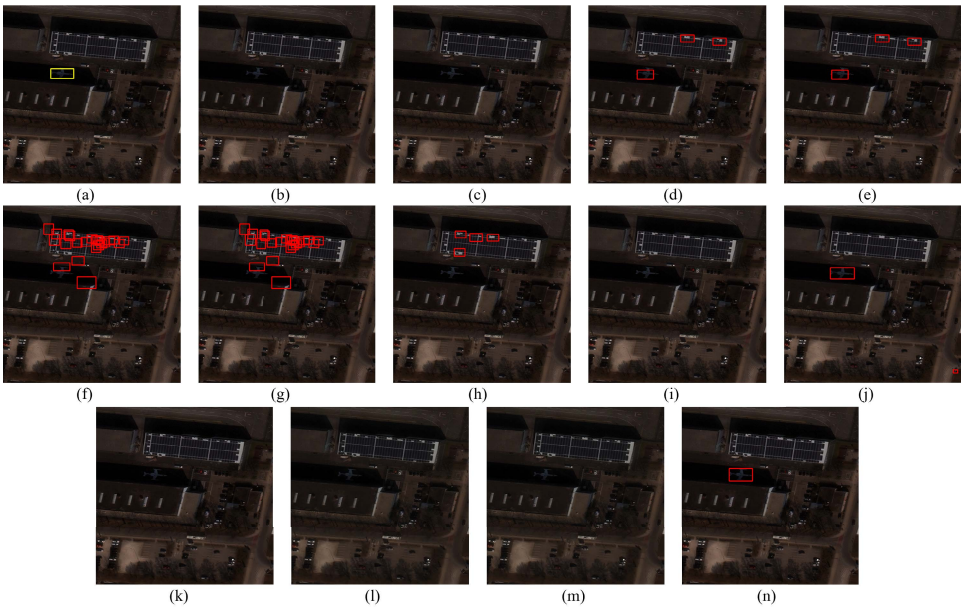


Fig. 11. Detection results of different object detection algorithms on example image 2 from the RarePlanes dataset (conf_thre = 0.5). (a) GT. (b) Faster RCNN. (c) RetinaNet. (d) Cascade RCNN. (e) FCOS. (f) DETR. (g) Sparse RCNN. (h) DINO. (i) Yolov8. (j) Yolov11. (k) DoubleFPN. (l) SpecDETR. (m) DAHNet. (n) PSG-HTD.

with surrounding ground equipment and in a dark building shadow with low target–background contrast, respectively. Most comparison methods exhibit missed detections, false positives, or incomplete target localization. Notably, SpecDETR misses the shadowed aircraft despite its slightly higher overall mAP@0.5:0.95. PSG-HTD accurately detects the aircraft in both images without additional false detections, suggesting that the generated hyperspectral representation and spatial–spectral feature extraction enhance target–background discrimination under challenging conditions.

2) *Validation Results on the RSOD-VHR10 Dataset:* Table III shows the accuracy comparison of different target detection algorithms on the RSOD-VHR10 dataset. PSG-HTD also achieved the highest precision, recall, and mAP@0.5, reaching 0.828, 0.689, and 0.714, respectively.

Figs. 12 and 13 compare detection results on two representative RSOD-VHR10 images, where yellow boxes indicate ground-truth labels and the confidence threshold is set to conf_thre = 0.5. The images contain multiple aircraft on an open apron and densely arranged around terminal buildings, respectively. Most comparison methods exhibit missed detections or false positives, and the three hyperspectral detectors also fail to reliably detect all targets. In particular, the aircraft at the far right of the second image is missed by all comparison

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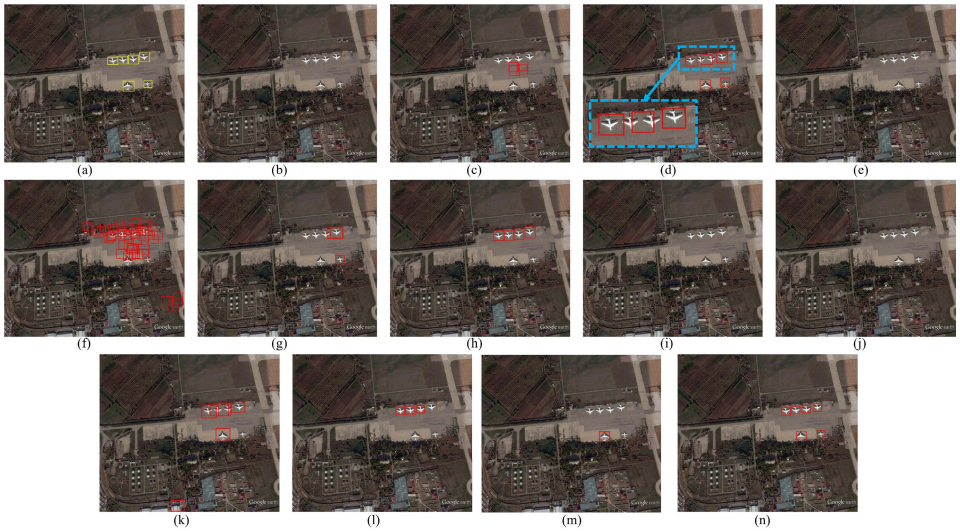


Fig. 12. Detection results of different object detection algorithms on example image 1 from the RSOD-VHR10 dataset (conf_thre = 0.5). (a) GT. (b) Faster RCNN. (c) RetinaNet. (d) Cascade RCNN. (e) FCOS. (f) DETR. (g) Sparse RCNN. (h) DINO. (i) Yolov8. (j) Yolov11. (k) DoubleFPN. (l) SpecDETR. (m) DAHNet. (n) PSG-HTD.

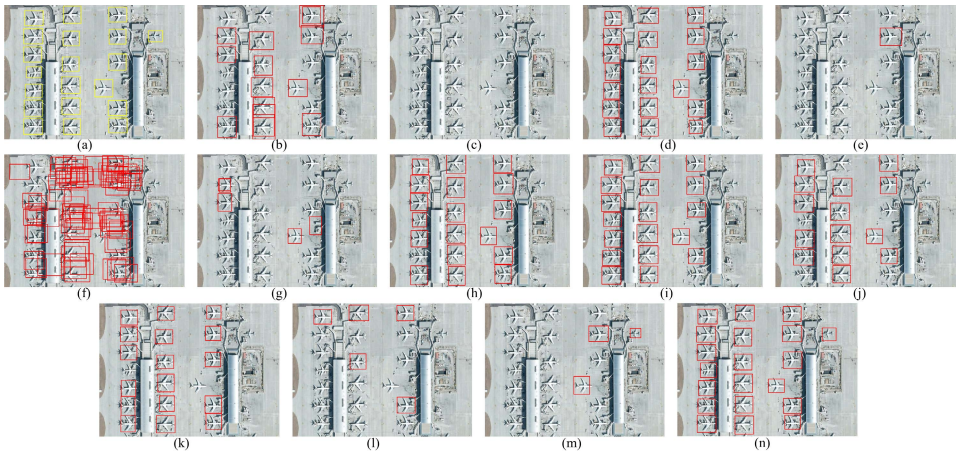


Fig. 13. Detection results of different object detection algorithms on example image 2 from the RSOD-VHR10 dataset (conf_thre = 0.5). (a) GT. (b) Faster RCNN. (c) RetinaNet. (d) Cascade RCNN. (e) FCOS. (f) DETR. (g) Sparse RCNN. (h) DINO. (i) Yolov8. (j) Yolov11. (k) DoubleFPN. (l) SpecDETR. (m) DAHNet. (n) PSG-HTD.

TABLE II

DETECTION RESULTS OF DIFFERENT OBJECT DETECTION ALGORITHMS ON THE RAREPLANES DATASET

Method	Precision	Recall	mAP@0.5	mAP@0.5:0.95
Faster RCNN	0.783	0.224	0.329	0.169
RetinaNet	0.382	0.418	0.474	0.187
Cascade RCNN	0.624	0.489	0.724	0.422
FCOS	0.836	0.413	0.565	0.255
DETR	0.175	0.393	0.515	0.190
Sparse RCNN	0.486	0.519	0.681	0.309
DINO	0.682	0.610	0.844	0.472
Yolov8	0.705	0.572	0.641	0.333
Yolov11	0.713	0.605	0.669	0.365
DoubleFPN	0.922	0.385	0.397	0.281
SpecDETR	0.837	0.718	0.813	**0.642**
DAHNet	0.781	0.585	0.677	0.374
PSG-HTD	**0.923**	**0.861**	**0.917**	0.589

TABLE III

DETECTION RESULTS OF DIFFERENT OBJECT DETECTION ALGORITHMS ON THE RSOD-VHR10 DATASET

Method	Precision	Recall	mAP@0.5	mAP@0.5:0.95
Faster RCNN	0.209	0.127	0.233	0.095
RetinaNet	0.132	0.127	0.029	0.007
Cascade RCNN	0.582	0.331	0.561	0.268
FCOS	0.443	0.182	0.275	0.088
DETR	0.043	0.063	0.008	0.002
Sparse RCNN	0.589	0.254	0.249	0.080
DINO	0.744	0.419	0.705	0.306
Yolov8	0.774	0.526	0.618	0.258
Yolov11	0.667	0.526	0.593	0.243
DoubleFPN	0.583	0.304	0.287	0.125
SpecDETR	0.730	0.426	0.689	**0.395**
DAHNet	0.606	0.421	0.449	0.168
PSG-HTD	**0.828**	**0.689**	**0.714**	0.319

methods. PSG-HTD correctly detects all annotated aircraft in both images without false positives, demonstrating stronger target–background discrimination in these challenging scenes.

D. Ablation Study

To further verify the improvement effect of the proposed physically constrained SG module on the performance of

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TABLE IV
ABLATION EXPERIMENT ON THE RAREPLANES DATASET

Method	SG	Precision	Recall	mAP@0.5	mAP@0.5:0.95
PSG-HTD	-	0.696	0.551	0.606	0.293
PSG-HTD	√	**0.923**	**0.861**	**0.917**	**0.589**

TABLE V
ABLATION EXPERIMENT ON THE RSOD-VHR10 DATASET

Method	SG	Precision	Recall	mAP@0.5	mAP@0.5:0.95
PSG-HTD	-	0.554	0.439	0.383	0.114
PSG-HTD	√	**0.828**	**0.689**	**0.714**	**0.319**

the cross-domain HTD algorithm, ablation experiments are conducted on the RarePlanes and RSOD-VHR10 datasets to compare the target detection performance before and after adding the SG module. The experimental results are shown in Table IV.

As shown in Table IV, removing the SG module leads to a notable drop in PSG-HTDs detection performance on the RarePlanes dataset (precision, recall, mAP@0.5, and mAP@0.5:0.95). In particular, mAP@0.5 and mAP@0.5:0.95 increase by 51.3% and 101.0%, respectively, when the SG module is included, demonstrating its effectiveness in improving detection accuracy for targets that are hard to distinguish using spatial features alone.

To further validate the effectiveness of the SG module, ablation experiments are also conducted on the RSOD-VHR10 dataset, with results summarized in Table V. Consistent with the RarePlanes results, the target detection accuracy of PSG-HTD on the RSOD-VHR10 dataset is significantly lower without the SG module. Specifically, mAP@0.5 and mAP@0.5:0.95 increase by 86.4% and 179.8%, respectively, after incorporating the SG module, further demonstrating its effectiveness for detecting targets that are difficult to identify using spatial features alone.

The ablation analysis further demonstrates that integrating the SG module improves the accuracy of cross-domain HTD while enhancing the discriminability between targets and background.

V. CONCLUSION

This article proposes a cross-domain HTD algorithm PSG-HTD based on reliable SG and dual-stream spatial–spectral fusion, to address insufficient spectral credibility and cross-domain spatial–spectral feature alignment. By incorporating spectral physical priors and the ADSSE module, PSG-HTD alleviates spectral distortion in multispectral-to-hyperspectral generation while improving target detection accuracy and generalization.

By integrating physical priors with deep learning, PSG-HTD provides a physically credible and generalizable solution for HTD under limited annotated samples. Future work will extend the framework to SG across different spectral scales and explore unified HTD across multisource sensors, scenarios, and resolutions.

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